

SEC P vs NP log 2026-05-16

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-16 21:44 UTC

SEC P vs NP — daily log 2026-05-16

4 cycles recorded

Layer-Commutator Frobenius Discrepancy Separates Read-Twice BP for IP₂

- **Verdict:** INCONCLUSIVE
- **Bridge:** Non-commutative operator discrepancy (Frobenius norm of layer-transition commutators, viewed as a Σ^b_1 -witnessing measure in bounded arithmetic à la Cook-Krajíček) $\times$ Read-twice branching programs computing the inner-product-mod-2 function IP₂
- **Recorded:** 2026-05-16 20:13 UTC
- **Entry ID:** 0e06b5f9fc52

Statement

Let P be a read-twice branching program with state set S ($|S|=s$) on n Boolean variables. For each of the $2n$ read-positions $r \in \{1, \dots, 2n\}$ let $A_r \in \mathbb{R}^{s \times s}$ be the averaged transition matrix $A_r := (T_r^0 + T_r^1)/2$, where T_r^b is the 0/1 matrix sending state u to state v if reading bit b at position r drives $u \rightarrow v$. Define the layer-commutator Frobenius discrepancy $\rho(P) := \log_2(1 + \sum_{1 \leq r < q \leq 2n} \|A_r A_q - A_q A_r\|_F^2 / s^2)$. Conjecture: (a) $\rho(P) \leq 6 \cdot \log_2(s) + 4$ for every read-twice BP P ; (b) for every read-twice BP P that computes IP₂ on $n \geq 4$ inputs, $\rho(P) \geq (1/8) \cdot n$. A single P violating either inequality refutes the conjecture; together they would force $\text{size}(P) \geq 2^{\{n/48\}}$ for read-twice BPs computing IP₂.

Rationale

Bounded-arithmetic witnessing of BP membership relies on local invariants that are Σ^b_1 -definable from the BP layer data; commutator energies of averaged layer matrices are exactly that kind of computable witness, while operator non-commutativity is the algebraic signature of the bilinear structure $\text{IP}_2(x,y) = \langle x, y \rangle_2$. Read-once BPs decompose into block-triangular layers where averaged transitions essentially commute on each reachable mass, but IP₂ forces every read-twice realization to mix x -reads and y -reads non-trivially. Because ρ is a Frobenius (L^2 spectral) quantity rather than a polynomial in the entries, the construction does not algebraize and the invariant is computed on BP structure (not the truth table), so it dodges Natural Proofs.

Novelty

- Judge: NOVEL over 8 arXiv hits

Top hits consulted: - [1709.08890v2] Partial matching width and its application to lower bounds for branching programs - [1007.1875v2] Lower Bounds for Quantum Oblivious Transfer - [0710.5926v2]

Mod 2 cohomology of 2-local finite groups of low rank - [2603.27466v1] English translation of Frobenius' and Stickelberger's "On the theory of elliptic functions" - [0906.1662v2] Frobenius morphisms of noncommutative blowups

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_12f9b311.py", line 211, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_12f9b311.py", line 148, in run_trial
    brs_bp = build_brs_bp(n)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_12f9b311.py", line 67, in build_brs_bp
    T0[u][v] = 1
    ~~~~~^^^
```

IndexError: list assignment index out of range

Judge reasoning

The test crashed with an IndexError in build_brs_bp before any trials produced data, so the pre-registered support condition cannot be evaluated. | next: Fix the indexing bug in build_brs_bp (state-set sizing for the BRS construction) and rerun the 150-trial sweep.

Matroid Matching of Term Co-Occurrence Graph Bounds k-CLIQUE Monotone DNF

- **Verdict:** INCONCLUSIVE
- **Bridge:** Matroid matching theory (Lovász 1980 polymatroid matching theorem, applied to term co-occurrence graphs of monotone DNFs) × Monotone DNF size and Karchmer-Wigderson protocols for the k-CLIQUE indicator
- **Recorded:** 2026-05-16 20:46 UTC
- **Entry ID:** 02c57a92c88c

Statement

For a monotone DNF $F = T_1 \vee \dots \vee T_m$ on n variables, build the *term co-occurrence graph* $H_F = ([n], E_F)$ with $E_F = \{\{u, v\} : \exists i, \{u, v\} \subseteq T_i\}$, and define $\mu(F) = \text{match}(H_F) / \lceil \log_2(1+m) \rceil$, where $\text{match}(H_F)$ is the size of a maximum matching in H_F . We conjecture: (i) μ is subadditive under DNF conjunction-expansion, $\mu(F \wedge G) \leq \mu(F) + \mu(G)$; (ii) $\mu(F) \leq 4$ for every monotone DNF F of 'log-width' type, namely $m \leq n^c$ and $\max_i |T_i| \leq c \cdot \log_2 n$ for any fixed $c \geq 1$; (iii) for the canonical k-CLIQUE DNF $F_{\{v, k\}}$ on v vertices with $k = \lceil \log_2 v \rceil$ (variables = the $C(v, 2)$ edges), $\mu(F_{\{v, k\}}) \geq v / (2 \cdot (\log_2 v)^2)$, i.e. unbounded as $v \rightarrow \infty$.

Rationale

Karchmer-Wigderson translates monotone formula size into communication on terms; matroid matching of the co-occurrence graph H_F is a representation-level polymatroid invariant capturing how 'expansive' the term overlap pattern is. Log-width poly-DNFs use few variables per term, keeping matchings tame, while every pair of vertices lies in some k-clique-term for k-CLIQUE, forcing H_F to be the complete graph K_v and matchings to scale linearly. Because μ reads the

syntactic term structure rather than the truth table, it sidesteps Razborov-Rudich naturalness while staying squarely inside the KW monotone-protocol techniques.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [1503.06447v1] Sum-of-squares lower bounds for planted clique - [2405.01069v1] Oriented Ramsey numbers of graded digraphs - [2505.06446v1] Structured Prediction with Abstention via the Lovász Hinge - [1102.2932v2] Applications of Monotone Rank to Complexity Theory

Empirical Test

- exit code: 1, elapsed: 0.36s

```

TRIAL: {'metric_name': 'mu', 'metric_value': 0.0, 'instances_tested': 9, 'conjecture_holds': False, '
CLIQUE bound failed: mu=0.0 < 0.4489652833957362 for v=6'}
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CLIQUE bound failed: mu=0.0 < 0.4489652833957362 for v=6'}
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CLIQUE bound failed: mu=0.0 < 0.4489652833957362 for v=6'}
TRIAL: {'metric_name': 'mu', 'metric_value': 0.0, 'instances_tested': 9, 'conjecture_holds': False, '
CLIQUE bound failed: mu=0.0 < 0.4489652833957362 for v=6'}
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CLIQUE bound failed: mu=0.0 < 0.4489652833957362 for v=6'}

```

--STDERR--

Traceback (most recent call last):

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_eb505197.py", line 137, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                        ~~~~~~
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_eb505197.py", line 137, in <genexpr>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                        ~~~~~~

```

KeyError: 'seed'

Judge reasoning

Safety rail: critic_challenge_falsified | original: The test reports conjecture_holds=False with an explicit counterexample: for the k-CLIQUE DNF at $v=6$, $\mu=0.0 < 0.449$, violating clause (iii) of the pre | next: Recompute $\mu(F_{\{v,k\}})$ by hand for small v to determine whether the lower bound $v/(2 \cdot (\log_2 v)^2)$ is fundamentally wrong or whether the matching/term-graph construction in the test mis-encodes the k-CLIQUE DNF.

Persistent H1 of Random Row Subclouds Bounds DISJ Communication

- **Verdict:** INCONCLUSIVE

- **Bridge:** Persistent homology / topological data analysis (Vietoris-Rips H_1 with total bar-length functional) $\times$ Randomized communication complexity of DISJOINTNESS (worst-case lower bound via sub-sampled row metric)
- **Recorded:** 2026-05-16 21:14 UTC
- **Entry ID:** 8fb2563d7c14

Statement

Let $M \in \{0,1\}^{\{N \times N\}}$ be the communication matrix of a Boolean function ($N = 2^n$). For a uniformly random k -element subset S of row indices with $k = \lceil \sqrt{N} \rceil$, view the rows $\{M[x, \cdot] : x \in S\}$ as a point cloud in $\{0,1\}^N$ under Hamming distance, and let $\tau_{\text{PH}}(M;S) := \sum_{(b,d) \in \text{Dgm}_1(\text{VR}(S))} (d - b)$ denote the total persistence of the 1-dimensional bars of the Vietoris-Rips filtration of that sub-cloud, with all distances normalised by N . Conjecture: there is an absolute constant $c > 0$ such that for every Boolean matrix M , $\text{CC}_R(M) \geq c \cdot \log_2(1 + E_S[\tau_{\text{PH}}(M;S)] \cdot k)$; moreover $E_S[\tau_{\text{PH}}(M_{\text{DISJ}_n};S)] = \Omega(1)$ for every $n \geq 4$, lifting the average-case topological signal to the worst-case bound $\text{CC}_R(\text{DISJ}_n) = \Omega(n)$.

Rationale

Persistent homology has been deployed in TDA but is essentially absent from communication-complexity lower bounds; unlike spectral or rank-based norms it is a multi-scale relational invariant (not a single-number truth-table statistic, so it dodges the natural-proofs trap) and does not algebrise because filtration birth/death values are not ring operations on a polynomial extension. The intuition is that high-CC matrices like DISJ scatter rows over many Hamming scales (producing long H_1 bars in subsamples) whereas low-CC matrices either cluster rows at few scales (e.g. Equality: all rows pairwise at Hamming distance 2, so every H_1 cycle is born and dies at the same filtration value, killing persistence) or admit small witnesses. Random-subset sampling matches the Impagliazzo-Wigderson-Hirahara avg-to-worst paradigm by deriving the worst-case CC_R bound from an expected topological invariant.

Novelty

- Judge: NOVEL over 11 arXiv hits

Top hits consulted: - [2301.07191v3] A New Construction of the Vietoris-Rips Complex - [2605.00705v1] Cohomological properties of the Vietoris-Rips Complex of a Hypercube Graph - [2001.07588v8] Vietoris-Rips Persistent Homology, Injective Metric Spaces, and The Filling Radius - [0104048v1] A study of the link between cosmic rays and clouds with a cloud chamber at the CERN PS - [0104076v1] CLOUD: an atmospheric research facility at CERN

Empirical Test

- exit code: 0, elapsed: 1.18s

```
eed": 23, "n": 6, "family": "DISJ", "metric_name": "log2(1 + E_S[\u03c4_PH] * k)", "metric_value": 12.
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TRIAL: {"seed": 23, "n": 6, "family": "EQ", "metric_name": "log2(1 + E_S[\u03c4_PH] * k)", "metric_val
TRIAL: {"seed": 23, "n": 6, "family": "RAND", "metric_name": "log2(1 + E_S[\u03c4_PH] * k)", "metric_v
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TRIAL: {"seed": 37, "n": 5, "family": "RAND", "metric_name": "log2(1 + E_S[\u03c4_PH] * k)", "metric_val
```

Judge reasoning

Pre-registered support condition is not met: EQ mean reaches 13.30 at $n=6$, vastly exceeding the ceiling $\log_2(6)+1 \approx 3.58$, and EQ matches IP exactly — clear evidence the metric is saturating at a k -determined ceiling rather than tracking communication complexity. The critic's challenge (small- n regime, trivially-satisfied $O(\log k)$ RHS, untested $\Omega(1)$ τ_{PH} claim) compounds this, so per the downgrade rule the apparent 100% support is not credible. | next: Re-run with a normalization that prevents me

Non-Backtracking Spectral Gap Bounds Tseitin Resolution Length

- **Verdict:** INCONCLUSIVE
- **Bridge:** Non-backtracking (Hashimoto) operator spectrum on graphs $\times$ Resolution refutation length / KW-refutation game depth for Tseitin formulas
- **Recorded:** 2026-05-16 21:44 UTC
- **Entry ID:** a58cbdc6e05

Statement

Let G be a connected d -regular graph ($d \geq 3$) on n vertices with odd-total Tseitin charge σ , let $B(G)$ be the $2m \times 2m$ Hashimoto non-backtracking edge operator with spectral radius $\rho(B)$, and define the graph invariant $\nu(G) := n \cdot \max(0, \log((d-1)/\rho(B)))$. We conjecture (i) every Resolution refutation π of $T(G, \sigma)$ satisfies $|\pi| \geq 2^{(c \cdot \nu(G))}$ for an absolute constant $c \geq 1/16$, and equivalently (via Pudlák's KW-refutation game) the refutation game has communication cost $\geq c \cdot \nu(G)$; (ii) $\nu(G) = O(1)$ for every graph that is not a vertex expander (any G with a balanced cut of size $o(n)$ has $\rho(B) \rightarrow d-1$, hence $\nu \rightarrow 0$). A single Tseitin instance refuted by Resolution in fewer than $2^{(c \cdot \nu(G))}$ steps refutes the conjecture.

Rationale

The Hashimoto/Ihara non-backtracking spectrum captures geometric-group-theoretic cogrowth and Ramanujan-style expansion finer than the ordinary Laplacian, yet is computed by a single polynomial-time eigen-decomposition. KW-refutation games convert Resolution refutations into edge-charge-flow games on G whose communication cost is bounded by precisely the non-backtracking mixing rate of charges along geodesics, so $\nu(G)$ is the natural information-flow witness for the BSW expansion-vs-width principle. The invariant is structural (graph-only) rather than truth-table single-number, sidesteps Natural Proofs, and uses no polynomial extension of the Boolean ring, so it avoids Algebrization.

Novelty

- Judge: NOVEL over 3 arXiv hits

Top hits consulted: - [2209.05839v3] On bounded depth proofs for Tseitin formulas on the grid; revisited - [2103.09609v1] Characterizing Tseitin-formulas with short regular resolution refutations - [2601.13117v2] The Case for Cardinality Lower Bounds

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b2cd758f.py", line 257, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b2cd758f.py", line 197, in run_trial
    B = build_non_backtracking_matrix(graph)
    ~~~~~^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b2cd758f.py", line 129, in build_non_ba
    if graph[u][j] != v:
    ~~~~~^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an `IndexError` in `build_non_backtracking_matrix` before any data was produced, so neither the support nor the falsification condition can be evaluated. | next: Fix the indexing bug in `build_non_backtracking_matrix` (likely an off-by-one or assumption about `graph[u]`'s neighbor list length) and re-run the pre-registered protocol.